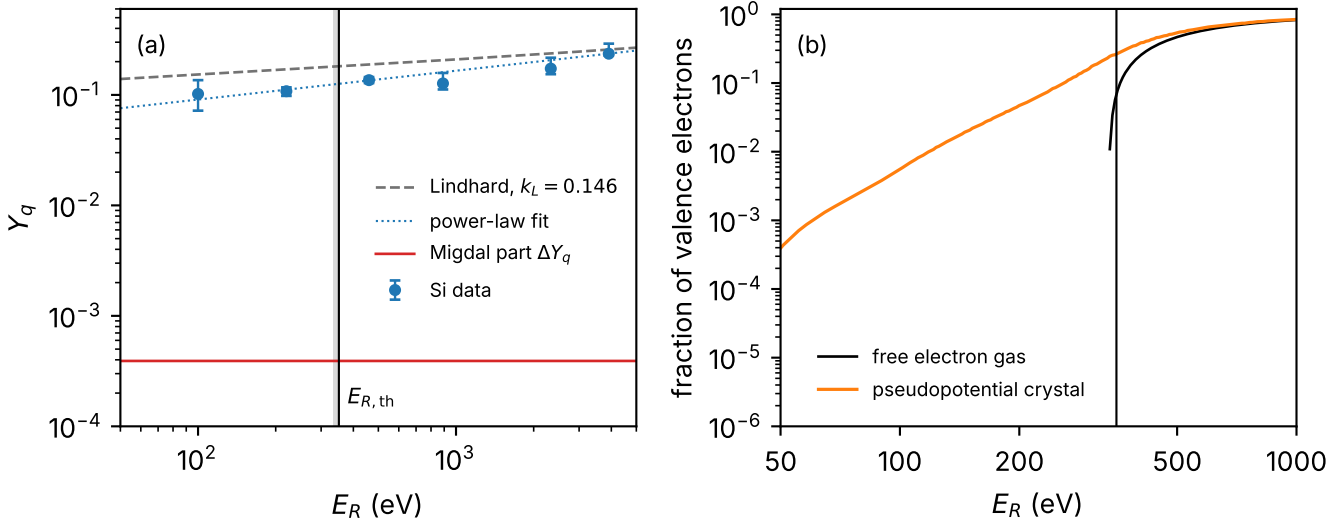
(a)
Y_q
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{2}
10^{3}
E_R (eV)
Lindhard, $k_L = 0.146$
power-law fit
Migdal part ΔY_q
Si data
$E_{R,\mathrm{th}}$
(b)
fraction of valence electrons
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
50
100
200
500
1000
E_R (eV)
free electron gas
pseudopotential crystal